Microeconomics

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Economics is bigger than this course

One shared question, different lenses: **what shapes wages?**

CLASSICAL

Value and distribution
Land rent; wages and profits
Surplus and accumulation

Pay: shares of total output.

MARXIAN

Labour and surplus value
Exploitation; accumulation
Class and ownership

Pay: class and bargaining.

NEOCLASSICAL

Consumer and firm
Risk; intertemporal choice
Topkis: monotone statics

THIS COURSE

General equilibrium
Games and information

Pay: productivity and scarcity.

AUSTRIAN

Market process
Dispersed knowledge
Entrepreneurship

Pay: discovery and coordination.

KEYNESIAN

Effective demand
Post-Keynesian uncertainty
Financial instability

Pay: demand and employment.

INSTITUTIONAL

Rules and organisations
Property rights
Transaction costs

Pay: institutions and bargaining.

BEHAVIOURAL

Bounded rationality
Prospect theory
Social preferences

Pay: norms and reference points.

EVOLUTIONARY

Routines and selection
Innovation
Path dependence

Pay: changing technologies.

Why do we need microeconomics?

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1. **Describe the world with explicit, reproducible models.** State assumptions, derive mechanisms and test predictions. Build structural models for real-world data and policy counterfactuals.
2. **Start from people and firms.** Build aggregate outcomes from their decisions. After the Lucas critique (1976), microfoundations became central to much of modern macroeconomics. Central banks and research institutes use these tools regularly.
3. **Make and challenge economic arguments in court.** In US antitrust and damages cases, economists write expert reports, build counterfactuals and scrutinise one another's evidence.

→ An Amazon pricing case: mechanism, evidence, counterfactual

The course at a glance

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- **7 lecture blocks** and **7 seminars**.
- **3 homeworks** with equally weighted assessed components: **20%**.
- **1 project**: an economic-puzzle essay and a short presentation: **10%**.
- **1 final exam**: **70%**.

Final grade $= 0.20\,HW + 0.10\,Project + 0.70\,Exam$.

Each homework has two parts

1. **A mock exam.** Exam-level difficulty and length. Try it honestly in one **120-minute sitting**. The TA will mark it as a real exam so that you can assess your readiness. Errors in this part do not reduce your final course grade.
2. **An assessed problem set.** Problems will, on average, be harder than exam questions. This part is marked in the usual way.

Write homework solutions by hand, on paper.

Discuss approaches, but submit your own reasoning. If you use generative AI, disclose the tool and its role and be ready to explain every step.

The project: Why does this happen?

1. Find a concrete economic phenomenon that puzzles you.
2. Say **who chooses**, what they want and what constrains them.
3. Explain the observation using one main economic mechanism.
4. Give an alternative explanation or a prediction that could distinguish mechanisms.
5. Use evidence and cite your sources.

Deliverables: a 600–900-word essay and a short presentation of the idea.

Topic after lecture 2; mechanism after lecture 4; final essay after lecture 6, before lecture 7. Presentations and discussion at seminar 7.

The final exam

- **Two hours**, in person, written and individual.
- **True / false statements with explanations.**
- **Problems** requiring a clear line of reasoning.
- **One qualitative economic question.**
- Explanations and economic interpretation matter, not just the final answer.

The date, detailed format and permitted materials will be announced in advance.

Office hours and questions

- **Office hours: by request.**
- Message me on **Telegram** to arrange a time or ask a question.
- Email: astolyarov@nes.ru.
- Ask during class when a definition, assumption or argument is unclear.

Seven lecture blocks: the topic map

	Lecture	Seminar / milestone
1	Preferences, choice, utility	Relations, indifference curves, budgets
2	Utility maximisation; Marshallian demand	Lagrangian, indirect utility, Roy
3	Duality, Slutsky, revealed preference	Hicksian demand, expenditure, WARP; HW 1
4	Choice under risk	Expected utility, risk aversion
5	The firm	Technology, profit, costs, duality; HW 2
6	Monotone comparative statics	Topkis, supermodularity
7	Intertemporal choice and finance	Saving, borrowing; project presentations; HW 3

Amazon and off-platform prices

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FTC and states v. Amazon (complaint filed in 2023).

- The complaint alleges that Amazon penalised sellers for offering lower prices on competing online stores.
- A proposed mechanism: sellers raise off-platform prices to avoid losing visibility or sales on Amazon.

A hypothetical seller: unit cost $\$60$; platform fees $\$20$ on Amazon and $\$10$ on eBay. Prices $\$100$ and $\$90$ give the same $\$20$ margin.

What happens if selling for $\$90$ elsewhere is penalised? Who gains and loses?

Could platform investment or free-riding concerns justify a restriction? What evidence would distinguish the explanations?

FTC: complaint and alleged anti-discounting practices

→ Back to why microeconomics

Today: from primitives to useful assumptions

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Part I: describing choice

- Preference relations
- Choice correspondences and WARP
- Utility representation
- Ordinal meaning and its limits

Part II: useful assumptions

- Continuity and indifference curves
- Monotonicity, LNS and convexity
- Separability
- Quasilinearity

To model a choice, we need to know...

What do people **want**?

What **can** they do?

Start with what they want.

What is the primitive?

The choice set X: fully specified alternatives.

- **Discrete:** $X = \{\text{basic}, \text{standard}, \text{pro}\}$.
- **Continuous:** $X = \mathbb{R}^L_+$, quantities of L commodities.
- **Discrete–continuous:** tea with m units of milk, or coffee with m units of milk:

$$X = \{\text{tea}, \text{coffee}\} \times \mathbb{R}_+.$$

- Quality, terms, timing and relevant context belong in the definition of an alternative.

One relation is enough to start

A weak preference relation is a subset $\succeq \subseteq X \times X$.

$$x \succeq y \quad \Longleftrightarrow \quad \text{“}x \text{ is at least as good as } y\text{.”}$$

It is a statement about a pair of alternatives, not a numerical score or an observed purchase. Some introductory textbooks also introduce $\sim$ or $\succ$ separately. Do we need them as additional primitives?

Can we define indifference using only $\succeq$?

We already know how to say "x is at least as good as y". How could we say "x and y are equally good"?

$$x \sim y \iff x \succeq y \text{ and } y \succeq x.$$

Both comparisons hold. This differs from incomparability, where neither comparison holds.

Can we define strict preference using only $\succeq$?

What extra condition makes “at least as good” become “strictly better”?

$$x \succ y \iff x \succeq y \text{ and not } y \succeq x.$$

A second independent preference primitive is unnecessary.

Does “not $x \succ y$” imply “$y \succ x$”?

No: $x \sim y$ is already a counterexample. Without completeness, incomparability is another.

What does rationality mean in a model?

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Economists love to assume **rational preferences**. What consistency would a model actually need?

Completeness: for every $x, y \in X$, $x \succeq y$ or $y \succeq x$, or both.

Transitivity: for every $x, y, z \in X$,

$$x \succeq y,\ y \succeq z \quad \Longrightarrow \quad x \succeq z.$$

Rational preferences = completeness + transitivity.

This does not require selfishness, perfect information or easy computation.

Same customer, two restaurant menus

Restaurant A
Steak
Duck confit with potato gratin
Cutlet with mashed potatoes
Choice: **steak**.

Restaurant B
Steak
Duck confit with potato gratin
Seven-course tasting menu (lobster ravioli, venison with truffle jus)
Choice: **duck confit**.

How could you explain the change? What would you need to hold fixed?

Suppose the shared dishes, prices and information are identical, and each observed choice is the **unique favourite**. Can one fixed ranking explain both?

Your turn: three relations

1. On $\mathbb{R}^2_+$: $x \succeq y \iff x_1 + x_2 \geq y_1 + y_2$.
2. On $\mathbb{R}^3$: $x \succeq y$ iff $x_i \geq y_i$ in **at least two of the three coordinates**.
3. On $\mathbb{R}^2_+$: $x \succeq y \iff x_1 \geq y_1$ and $x_2 \geq y_2$.

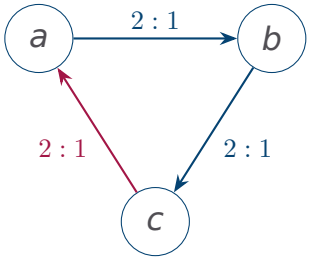
Check completeness and transitivity. Give a counterexample when a property fails.

For relation 2, try $(3, 2, 1)$, $(2, 1, 3)$ and $(1, 3, 2)$.

Condorcet's paradox: rational voters, cyclic majority

Each voter has a complete and transitive ranking. Compare alternatives by simple majority.

Voter	Individual ranking
1	$a \succ_1 b \succ_1 c$
2	$c \succ_2 a \succ_2 b$
3	$b \succ_3 c \succ_3 a$



Individual rationality need not produce a transitive collective ranking. No alternative beats every other alternative by majority.

We see choices, not the whole preference relation

A theory of preferences is useful, but we do not observe everybody's complete ranking. Usually we see choices from particular menus.

$$C(\{A, B, C\}) = \{A\}, \qquad C(\{B, C\}) = \{B\}.$$

These choices are consistent with $A \succ B \succ C$: the menu changed, while preferences need not have changed.

What must we observe besides a purchase before inferring anything about preferences?

Choice as the primitive

Let $\mathcal{D} = 2^X \setminus \{\varnothing\}$ be the collection of all nonempty menus.

Choice correspondence

$$C : \mathcal{D} \to 2^X, \qquad \varnothing \neq C(A) \subseteq A.$$

$C(A)$ records **all selected best alternatives**, not just one realised purchase.

A preference relation induces

$$C(A; \succeq) = \{x \in A : x \succeq y \text{ for every } y \in A\}.$$

A best element need not exist on an infinite menu. Nonemptiness is an additional requirement.

Consistency across menus: WARP

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Weak Axiom of Revealed Preference

For any menus A, B and any $x, y \in A \cap B$,

$$x \in C(A), \quad y \in C(B)$$
$$\Rightarrow \quad x \in C(B), \quad y \in C(A).$$

- Both alternatives must be available in both menus.
- One cannot be strictly beaten in one menu and selected over its rival in another.
- This formulation concerns the **whole correspondence**, including ties.

A two-menu inconsistency

Menu $A = \{a, b, c\}$

$$C(A) = \{a\}.$$

a is selected; b is available but not selected.

Menu $B = \{a, b, d\}$

$$C(B) = \{b\}.$$

b is selected; a is available but not selected.

Can a single complete and transitive preference relation generate both choice sets?

Preferences and choice: an equivalence

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Finite-set theorem

Let X be finite and C be nonempty on **every** nonempty subset of X.
Then C satisfies WARP if and only if there exists a complete, transitive relation $\succeq$ such that

$$C(A) = C(A; \succeq) \quad \text{for every nonempty } A \subseteq X.$$

- Preferences $\rightarrow$ choices: select every best alternative.
- Choices $\rightarrow$ preferences: infer $x \succeq y$ from $x \in C(\{x, y\})$.

$\rightarrow$ Proof, if time permits $\rightarrow$ Preferences $\rightarrow$ WARP

What choice data do not automatically reveal

1. We often observe one realised choice, not **all** best alternatives.
2. We observe only a sample of menus, not every subset of X.
3. A consistency test on that sample need not establish global rationalisability.

Observations, tie-breaking assumptions and the menu domain jointly determine what can be inferred.

If we only know that an observed choice is optimal, universal indifference can rationalise any such observations. Additional structure matters.

Utility is a representation

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Definition

$u : X \to \mathbb{R}$ represents $\succeq$ if, for every $x, y \in X$,

$$x \succeq y \quad \iff \quad u(x) \geq u(y).$$

Then

$$C(A; \succeq) = \arg\max_{x \in A} u(x).$$

- The numbers reproduce the entire ranking, including indifference.
- Representation turns choice into an optimisation problem.

Representation on a finite set

Complete and transitive preferences on a finite X admit a utility representation.

1. Partition alternatives into indifference classes.
2. Order the classes from worst to best.
3. Assign increasing numerical labels.

$$\underbrace{\{C, D\}}_{u=0} \prec \underbrace{\{B\}}_{u=1} \prec \underbrace{\{A, E\}}_{u=2}.$$

The ranking matters. The gaps between the labels do not.

Ordinal, not a happiness meter

Alternative	u	v
A	3	100
B	2	7
C	1	-20

Both functions represent $A \succ B \succ C$.

Increasing transformations preserve the ranking

If f is strictly increasing on $u(X)$, then $v = f \circ u$ represents the same preferences:

$$u(x) \geq u(y) \iff f(u(x)) \geq f(u(y)).$$

Examples: $u + 5$, $30u$, e^u. Squaring works on a nonnegative range, not in general. Why is a **weakly** increasing transformation not enough?

No interpersonal comparisons for free

Yaromir and Zoya both prefer x to y.

- We may rescale Yaromir's utility without changing any of his choices.
- We may independently rescale Zoya's utility without changing her choices.
- The sum $u_Y + u_Z$ can change its ranking of social outcomes after those rescalings.

An individual ordinal representation does not, by itself, supply interpersonal units or a social welfare criterion.

Does every ranking have a utility representation?

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Can every complete and transitive preference relation be represented by a real-valued utility function?

On $X = [0,1]^2$, compare the first coordinate first, and the second only to break ties:

$$x \succeq_{\text{lex}} y \iff [x_1 > y_1] \text{ or } [x_1 = y_1 \text{ and } x_2 \geq y_2].$$

- Complete and transitive.
- Any gain in coordinate 1 outweighs any loss in coordinate 2.
- No real-valued utility function represents this relation on $[0,1]^2$.

The non-representation proof uses uncountably many disjoint intervals; see the appendix.

Checkpoint

1. Is indifference a violation of completeness?
2. Can a utility representation generate intransitive preferences?
3. Does a different realised purchase prove that preferences changed?
4. What changes when we move from finite to infinite choice sets?

Continuity: comparisons survive limits

On $X \subseteq \mathbb{R}^L$, preferences are continuous if

$$x^n \to x, \quad y^n \to y, \quad x^n \succeq y^n \ \forall n$$
$$\Rightarrow \quad x \succeq y.$$

A useful representation theorem

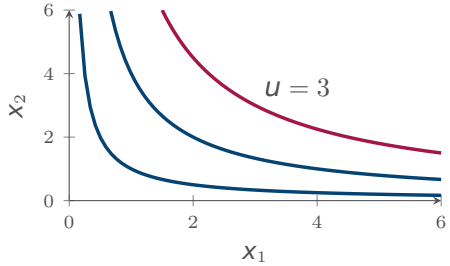
On $\mathbb{R}^L_+$, complete, transitive and continuous preferences admit a **continuous** utility representation.

Lexicographic counterexample: $(1/n, 0) \succ_{\text{lex}} (0, 1)$ for all n, but $(0, 0) \prec_{\text{lex}} (0, 1)$ in the limit.

Indifference curves: equally good bundles

For a given bundle $\bar{x}$, its indifference set is

$$I(\bar{x}) = \{x : x \sim \bar{x}\} = \{x : u(x) = u(\bar{x})\}.$$



These are level sets for the example $u(x) = \sqrt{x_1 x_2}$.
Different utility labels can describe the same indifference curves.

Monotonicity: fix the notation first

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Write $x \geq y$ if $x_i \geq y_i$ for every i, and $x \gg y$ if $x_i > y_i$ for every i.

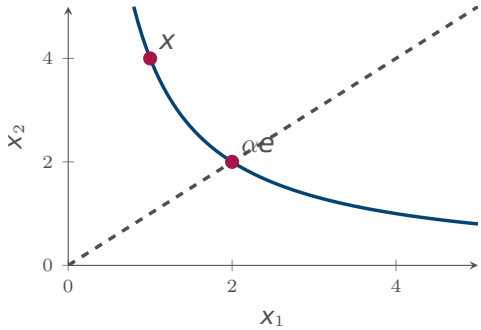
Two useful conditions

Monotonicity: $x \gg y \Rightarrow x \succ y$.

Strong monotonicity: $x \geq y,\ x \neq y \Rightarrow x \succ y$.

- More of **every** good versus more of **at least one** good, with no losses.
- Neither is part of the definition of rationality.
- The assumption is inappropriate for a bad unless we redefine the commodity.

A constructive picture: the diagonal bundle



Now add monotonicity to rationality and continuity. Each x is indifferent to a unique diagonal bundle αe, where $e = (1, \ldots, 1)$.

$$u(x) = \alpha \quad \text{if } x \sim \alpha e.$$

A useful way to **construct** a utility scale.

→ Proof under monotonicity

Local nonsatiation: a better bundle nearby

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Coffee and cake: could there be a bundle where every sufficiently small change makes you no better off?

For every $x \in X$ and every $\varepsilon > 0$, there is $y \in X$ such that

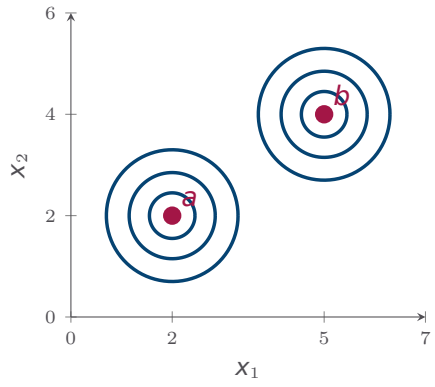
$$\|y - x\| < \varepsilon, \qquad y \succ x.$$

- There is no local maximum of any representing utility function.
- On $\mathbb{R}^L_+$: strong monotonicity $\Rightarrow$ monotonicity $\Rightarrow$ LNS.
- The converse implications need not hold.

Example

$u(x_1, x_2) = x_1$ is locally nonsatiated but not strongly monotone.

Two bliss points



Can these preferences be rational and continuous? Locally nonsatiated? Convex?

Convex preferences: mixing bundles

Assume X is convex. Preferences are **convex** if, for every $\lambda \in [0,1]$,

$$x \succeq z,\ y \succeq z \quad \Rightarrow \quad \lambda x + (1-\lambda) y \succeq z.$$

Equivalently, every upper contour set $\{x : x \succeq z\}$ is convex.

Preferences are **strictly convex** if, for distinct x, y and $0 < \lambda < 1$, the conclusion is strict: $\lambda x + (1-\lambda) y \succ z$.

In particular, mixing two distinct indifferent bundles is weakly better under convexity and strictly better under strict convexity.

This is a mixture of quantities, not a lottery over the two bundles.

Convex functions and convex preferences

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On a convex domain, u is a **convex function** if

$$u(\lambda x + (1-\lambda)y) \leq \lambda u(x) + (1-\lambda)u(y), \qquad \lambda \in [0,1].$$

Do convex preferences imply a convex utility function? Must a representing utility be convex?

First example: $u(x) = x_1 + x_2$. It is affine: both convex and concave. The preferences are convex.

Increasing transformation: $v(x) = e^{x_1+x_2}$. It represents the same preferences; it is convex but not concave.

These two examples do not establish the general claim. Which property of utility actually captures convex preferences?

Convex preferences and quasiconcavity

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A convex representation need not exist. For $u(x) = \sqrt{x_1 x_2}$,

$$a = (1, 4) \sim b = (4, 1), \qquad m = \tfrac{1}{2}(a + b) = (2.5, 2.5) \succ a.$$

Any convex representation g would require

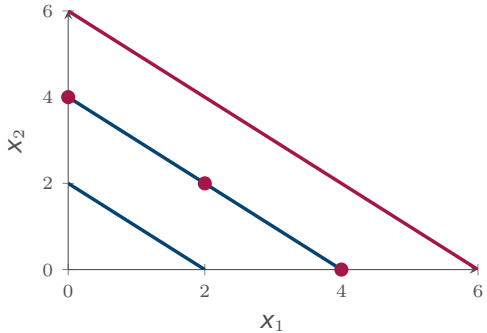
$$g(m) \leq \tfrac{1}{2} g(a) + \tfrac{1}{2} g(b) = g(a),$$

contradicting $m \succ a$.
The right property is **quasiconcavity**:

$$u(\lambda x + (1 - \lambda) y) \geq \min\{u(x), u(y)\}.$$

Convex preferences $\iff$ quasiconcavity of a representing utility. This property survives strictly increasing transformations.

Perfect substitutes: a straight-line ranking

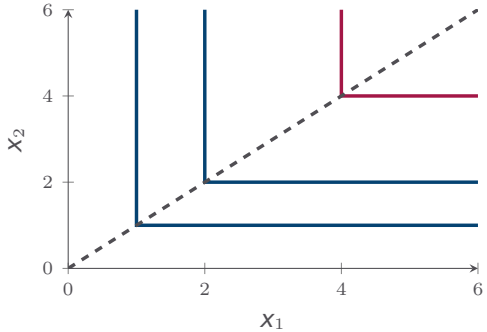


$$u(x) = x_1 + x_2.$$

- Strongly monotone
- Convex, not strictly convex
- $(4, 0) \sim (0, 4) \sim (2, 2)$

One unit of either good contributes the same amount to the ranking.

Perfect complements: useful in fixed proportions

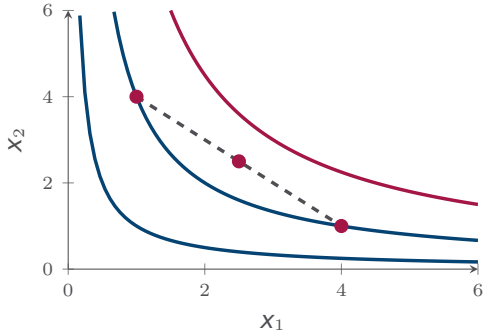


$$u(x) = \min\{x_1, x_2\}.$$

- Monotone, not strongly monotone
- Locally nonsatiated
- Convex, not strictly convex

$$(2, 2) \sim (5, 2).$$

Cobb–Douglas: mixing can help



$$u(x) = \sqrt{x_1 x_2}.$$

$u(1,4) = u(4,1) = 2$, but

$$u(2.5, 2.5) = 2.5.$$

On $\mathbb{R}^2_{++}$: strongly monotone and strictly convex preferences.
On $\mathbb{R}^2_{+}$, axis bundles have zero utility; the strict claims fail at the boundary.

Separability: a ranking inside a larger bundle

Let $X = Y \times Z$. Think of Y as food and Z as housing and transport.

Preferences over Y do not depend on Z

For all $y, y' \in Y$ and $z, z' \in Z$,

$$(y, z) \succeq (y', z) \quad \Longleftrightarrow \quad (y, z') \succeq (y', z').$$

- Your ranking of two food bundles does not depend on which car you have.
- Your **food budget** can still depend on car spending.
- The condition is not symmetric between Y and Z.

Separability: summarise a group of goods

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Let $y = (y_1, y_2)$ be a food bundle and z everything else. Suppose the ranking of food bundles does not depend on z.

Summarise the food bundle by one number $v(y)$, then combine it with z:

$$u(y, z) = U(v(y), z).$$

For each fixed z, U must increase strictly with the food index: a better food bundle must still be better.

Example:

$$v(y) = \sqrt{y_1 y_2}, \qquad U(v, z) = v + \log(1 + z).$$

$(1, 4)$ has food index 2; $(1, 1)$ has food index 1.

At **any fixed** z, the first food bundle ranks higher. This does not fix the food budget. → Formal construction

Quasilinearity: a numeraire plus a valuation

$$u(a, y) = a + v(y), \qquad a \geq 0.$$

- a is money left over, or another **numeraire** good.
- $v(y)$ measures the valuation of y in numeraire units.
- Adding the same feasible amount t to both numeraire holdings preserves their ranking.

$$(a, y) \succeq (a', y') \iff (a + t, y) \succeq (a' + t, y').$$

A convenient assumption with economic content: money compensation is independent of the starting numeraire level.

No income effects: mind the boundary

Let $u(a, y) = a + 4\sqrt{y}$, with both prices equal to 1 and $a + y \leq m$.
Substitute the exhausted budget:

$$\max_{0 \leq y \leq m} \; m - y + 4\sqrt{y}.$$

Income m	y^*	a^*
8	4	4
12	4	8
2	2	0

The unconstrained best level is $y = 4$.
Thus $y^* = \min\{m, 4\}$. Demand for y is income-independent when the numeraire constraint is slack; income matters when that constraint binds.

What can they do? The budget set

For $p_1, p_2 > 0$ and income $m > 0$,

$$B(p, m) = \{x \in \mathbb{R}^2_+ : p_1 x_1 + p_2 x_2 \leq m\}.$$

The budget line is $x_2 = m/p_2 - (p_1/p_2)x_1$. Its intercepts are m/p_1 and m/p_2; its slope is $-p_1/p_2$.

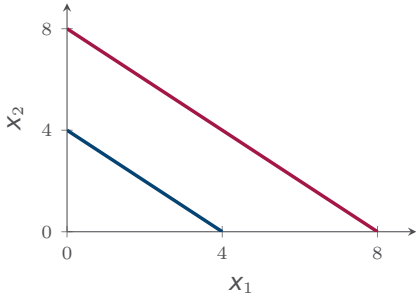
Higher income at fixed prices: a parallel outward shift.

A higher p_1 at fixed p_2, m: a rotation inwards around the vertical intercept.

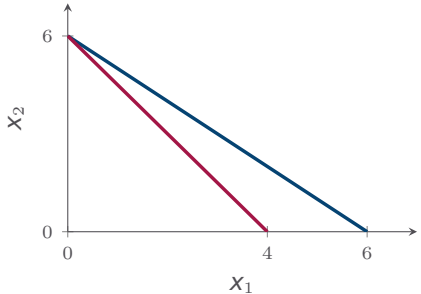
Preferences describe wants; the budget describes possibilities. Next lecture: solve the consumer's optimisation problem.

Budget geometry: income versus a price change

Higher income, fixed prices

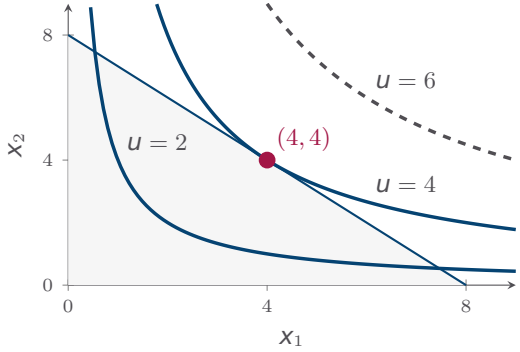


Higher p_1, fixed p_2, m



The feasible set expands in the first case and shrinks in the second. Neither change is itself a change in preferences.

Wants and possibilities in one picture



Possibilities:

$p_1 = p_2 = 1$, $m = 8$.

$x_1 + x_2 \leq 8$.

Wants:

$u(x) = \sqrt{x_1 x_2}$.

The best affordable bundle is $(4, 4)$, with utility 4.

Why is $u = 6$ unattainable? Why settle for $u = 2$?

Exit ticket

1. Why can a strictly increasing transformation change concavity but not preferences?
2. Give rational preferences that are not strongly monotone.
3. Does convexity imply that every mixture is strictly better?
4. Does separability imply that spending on Y is independent of income?
5. What is the boundary caveat for quasilinear demand?

Use a definition, example or short argument for each answer.

What to keep, and what to read

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Main reading

Hal R. Varian, *Intermediate Microeconomics: A Modern Approach*, **9th edition** (2014).

- Chapter 2: *Budget Constraint*.
- Chapter 3: *Preferences*.
- Chapter 4: *Utility*.

Optional advanced reading

Mas-Colell, Whinston and Green, *Microeconomic Theory* (1995).

Chapter 1: *Preference and Choice*; Chapter 3: *Classical Demand Theory*, sections 3.B–3.C on preferences and utility.

Appendix: why rational preferences imply WARP

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Suppose $x, y \in A \cap B$, $x \in C(A)$ and $y \in C(B)$.

1. Because $y \in A$, optimality of x gives $x \succeq y$.
2. Because $x \in B$, optimality of y gives $y \succeq x$.
3. For every $z \in B$, $y \succeq z$; transitivity gives $x \succeq z$.
4. Therefore $x \in C(B)$. The symmetric argument gives $y \in C(A)$.

The argument uses transitivity and the full best-alternative interpretation of C. Finite menus ensure nonemptiness separately.

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Appendix: reconstructing the relation from choice

NES

Define $x \succeq y$ iff $x \in C(\{x, y\})$.

- **Completeness:** pair menus have nonempty chosen sets.
- **Transitivity:** let $x \succeq y \succeq z$ and choose $w \in C(\{x, y, z\})$. WARP with the relevant pair menus first brings y or x into the triple, and then brings x into it. WARP with $\{x, z\}$ gives $x \succeq z$.
- **Representation of choice:** if $x \in C(A)$, WARP with each $\{x, y\}$ yields $x \succeq y$. Conversely, if x beats every $y \in A$, choose $y \in C(A)$ and use WARP with $\{x, y\}$ to obtain $x \in C(A)$.

Domain: all nonempty subsets of a finite X. Repeated alternatives in a triple give trivial cases.

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Appendix: limited data can hide a cycle

Suppose only pair menus are observed:

$$C(\{s,p\}) = \{s\}, \qquad C(\{p,r\}) = \{p\}, \qquad C(\{s,r\}) = \{r\}.$$

- Different observed menus share only one alternative.
- WARP has no nontrivial two-alternative comparison across these menus.
- Yet the singleton chosen sets require $s \succ p \succ r \succ s$.
- No complete, transitive preference relation generates these full chosen sets.

Passing WARP on an arbitrary observed domain does not imply rationalisability on the full domain.

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Appendix: why lexicographic utility does not exist

Suppose a real-valued u represents lexicographic preferences on $[0,1]^2$.

1. For each $t \in [0,1]$, $u(t,1) > u(t,0)$.
2. Thus $I_t = (u(t,0), u(t,1))$ is a nonempty open interval.
3. If $s > t$, then $u(s,0) > u(t,1)$, so I_s and I_t are disjoint.
4. Each I_t contains a rational number; distinct intervals contain distinct selected rationals.

There cannot be an injection from uncountably many $t \in [0,1]$ into the countable rationals. Contradiction.

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Appendix: construction under monotonicity

Assume rationality, continuity and monotonicity on $\mathbb{R}^L_+$; let $e = (1, \dots, 1)$.

1. Continuity and monotonicity imply $x \geq y \Rightarrow x \succeq y$.
2. On $[0, M]$, $M = \max_i x_i$, the sets $\{a : ae \succeq x\}$ and $\{a : x \succeq ae\}$ are nonempty, closed and cover the interval. Connectedness makes them intersect.
3. If $a > b$, $ae \succ be$; hence the indifferent coefficient is unique. Set $u(x) = a$.
4. Transitivity and diagonal monotonicity give $x \succeq y \iff u(x) \geq u(y)$.
5. If $x^n \to x$, the coefficients $u(x^n)$ are bounded. Every convergent subsequence has limit $u(x)$ by continuity and uniqueness. Therefore $u(x^n) \to u(x)$.

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Appendix: separability in one construction

Fix z_0 and let $v(y) = u(y, z_0)$. For $t \in v(Y)$, pick $\hat{y}(t)$ with $v(\hat{y}(t)) = t$, and define

$$U(t, z) = u(\hat{y}(t), z).$$

- If two representatives have the same t, they are indifferent at z_0; independence makes them indifferent at every z. So U is well-defined.
- Since $\hat{y}(v(y))$ and y are indifferent at every z,

$$U(v(y), z) = u(y, z).$$

- If $t > t'$, strict preference at z_0 carries to every z, so $U(t, z) > U(t', z)$.

The constructed aggregator has domain $v(Y) \times Z$.

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Appendix: primitives behind quasilinearity

On $\mathbb{R}_+ \times Y$, assume rationality, a worst $\bar{y}$, and:

1. $(a, \bar{y}) \succeq (a', \bar{y}) \iff a \geq a'$.
2. For every y, there exists $t \geq 0$ with $(0, y) \sim (t, \bar{y})$.
3. Adding the same feasible amount to both numeraire holdings preserves comparisons in both directions.

Define $v(y)$ by $(0, y) \sim (v(y), \bar{y})$. Translation invariance gives

$$(a, y) \sim (a + v(y), \bar{y}).$$

Transitivity and condition 1 then imply

$$(a, y) \succeq (a', y') \iff a + v(y) \geq a' + v(y').$$

So $u(a, y) = a + v(y)$ represents preferences.

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Appendix: LNS and budget exhaustion

NES

Suppose $p \gg 0$, x^* is optimal and preferences are locally nonsatiated.
If $p \cdot x^* < m$, let $\delta = m - p \cdot x^* > 0$.
Whenever $y \in X$ satisfies $\|y - x^*\| < \delta/\|p\|$,

$$p \cdot y \leq p \cdot x^* + \|p\|\|y - x^*\| < m.$$

LNS gives a strictly better y within this neighbourhood. It is feasible, contradicting optimality.

LNS implies budget exhaustion at an existing optimum; it does not establish existence.

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Appendix: increasing transformations and risk

Compare $A = 2$ for sure with $B = 0$ or 3, each with probability $1/2$.

$$\mathbb{E}[A] = 2 > 1.5 = \mathbb{E}[B].$$

After the strictly increasing transformation $f(t) = e^t$,

$$\mathbb{E}[e^A] = e^2 \approx 7.39 < \frac{1 + e^3}{2} \approx 10.54 = \mathbb{E}[e^B].$$

The ranking of individual outcomes is unchanged, but the ranking of lotteries by expected transformed numbers changes. Ordinal deterministic utility and expected utility require different care.

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Appendix: answers to the relation exercise

NES

1. Sum of coordinates: complete and transitive, by the ordering of real numbers.
2. At least two of three coordinates: complete, but not transitive. Let $a = (3, 2, 1)$, $b = (2, 1, 3)$, $c = (1, 3, 2)$. Then $a \succ b$, $b \succ c$, $c \succ a$.
3. Componentwise comparison: transitive, but incomplete. $(2, 1)$ and $(1, 2)$ are incomparable.

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